

Chapter 2

Mechanics of futures markets

Full explanation with definitions, worked examples, and revision tables

John C. Hull • Ninth Global Edition

Based on the supplied Chapter 2 explanation, covering Sections 2.1–2.11, both business snapshots, the statistical idea in Question 2.35, and the final terminology comparisons.

Independent study notes. The chapter's historical regulatory and tax references are retained as learning context.

Contents

Before Section 2.1: the essential vocabulary	3
2.1 - Background: opening and closing a futures position	4
2.2 - Specification of a futures contract	5
2.3 - Convergence of futures and spot prices	8
2.4 - The operation of margin accounts	9
2.5 - OTC markets, collateral, and central clearing	14
2.6 - Market quotes	16
2.7 - Delivery	18
2.8 - Types of traders and types of orders	20
2.9 - Regulation	22
2.10 - Accounting and tax	23
2.11 - Forward versus futures contracts	25
The extra statistical idea in Question 2.35	26
The distinctions you should be able to explain aloud	28

Chapter 2: how futures markets operate

Chapter 2 explains how futures markets actually operate: how a trade is created, how you close it, who pays whom each day, what margin means, how delivery works, and how clearing reduces the risk of someone failing to pay.

This follows **Sections 2.1-2.11 of your uploaded Hull, 9th Global Edition**, including the two business snapshots. The numerical examples below are either identified as Hull's examples or are simplified examples for understanding.

Keep this question in mind throughout:

If two strangers agree today on a transaction that will happen months later, how does the market make sure their obligations are fulfilled?

Before Section 2.1: the essential vocabulary

Suppose you enter a contract involving **100 ounces of gold for December**.

Term	Meaning in this example
Underlying asset	Gold - the asset the contract refers to
Spot price	Price of gold for almost immediate delivery
Futures price	Current quoted price for the December futures contract
Contract size	100 ounces per contract
Long position	You take the buying side
Short position	You take the selling side
Delivery month	December
Counterparty	The party legally facing you under the contract
Position	Your outstanding contractual exposure
Settlement	Fulfilling payment or delivery obligations
P&L	Profit and loss

For ordinary commodity futures:

- **A long gains when the futures price rises.**
- **A short gains when the futures price falls.**

“Buying futures” does **not** mean that you immediately receive the commodity or pay its full purchase price.

2.1 - Background: opening and closing a futures position

How a futures trade begins

Hull uses a corn example:

- One contract represents **5,000 bushels**.
- A bushel is a quantity measure used in grain markets.
- One trader wants to buy September corn futures.
- Another wants to sell September corn futures.
- They agree at **600 cents per bushel**, meaning **\$6 per bushel**.

The reference amount is:

$$5,000 \times \$6 = \$30,000$$

The buyer becomes **long one September contract**. The seller becomes **short one September contract**.

The \$30,000 is the contract's underlying reference amount. It is not necessarily the cash deposited when the trade opens.

Who determines the futures price?

Buyers and sellers determine it through trading.

If buying interest becomes more aggressive, prices tend to rise until sellers are willing to transact. If selling becomes more aggressive, prices tend to fall.

Every completed contract still has one long and one short. Saying “more buyers than sellers” informally means stronger buying pressure, not contracts without matching sellers.

Open outcry and electronic trading

Open outcry is the historical system where traders physically meet, shout prices, and use hand signals.

Electronic trading uses computer systems to match orders.

The contractual obligation exists regardless of how the order was matched.

Closing out a position

Closing out, or **offsetting**, means entering the opposite trade in the **same futures contract**.

Existing position	Trade that closes it
Long three September corn contracts	Sell three September corn contracts
Short three September corn contracts	Buy three September corn contracts

You do not need to find your original trading partner.

Example

You buy one contract at \$6 per bushel and close it by selling at \$6.20.

$$\text{Profit} = 5,000(6.20 - 6.00) = \$1,000$$

The general formula is:

$$\text{Long futures P\&L} = NQ(F_{\text{exit}} - F_{\text{entry}})$$

where:

- **N**: number of contracts.
- **Q**: units per contract.
- **F**: quoted price per unit.

For a short, reverse the price difference.

A different delivery month does not close the position

If you buy September corn and sell December corn, you still have two exposures.

This is a **calendar spread**: opposite positions in different maturities.

Rolling a position means closing one maturity and opening another. It does not erase earlier gains or losses.

Business Snapshot 2.1: accidental cattle delivery

Hull tells a story, explicitly described as possibly apocryphal, about an employee who needed to close a client's long cattle position.

The employee **bought another contract instead of selling**.

The institution ended up long two contracts and had to handle delivery, including housing and feeding cattle until resale.

The lesson: direction and delivery deadlines are real operational details. A simple order mistake can create physical obligations.

2.2 - Specification of a futures contract

An exchange must define exactly what its contract means.

"I will deliver some wheat later" is too vague to trade reliably.

1. The underlying asset and its grade

A **grade** is an agreed quality classification.

Commodity quality affects value. Different wheat, corn, or orange-juice products are not automatically interchangeable.

Real-world example

A food manufacturer needs a specific quality of wheat. Without a quality rule, a futures seller might deliver very low-quality wheat while claiming it satisfies the contract.

The exchange therefore specifies acceptable grades.

Some contracts allow alternative grades with **premiums or discounts**:

- Better grade: seller receives an additional amount.
- Lower permitted grade: seller receives less.

These adjustments make different deliveries economically comparable.

2. Financial assets can also require detailed specifications

Currencies are relatively unambiguous: one Japanese yen is one Japanese yen.

Bond futures can be more complicated because several bonds may be eligible for delivery.

Relevant terms include:

- **Face value:** principal amount the bond promises to repay.
- **Coupon:** contractual interest payment.
- **Maturity:** date the principal is due.
- **Deliverable basket:** set of securities eligible for delivery.

Two bonds with different coupons and maturity dates can have different values. The futures contract therefore needs adjustment rules.

3. Contract size

Contract size is the quantity represented by one contract.

Examples used in the chapter include:

- Gold: 100 ounces.
- Corn: 5,000 bushels.
- Live cattle: 40,000 pounds.

These are the edition's examples; actual specifications must be checked before trading.

Why does size matter?

- **Too large:** small businesses cannot hedge precisely.
- **Too small:** many contracts are required, increasing transaction costs.

Example

A farmer expects to sell 120,000 pounds of cattle.

With 40,000 pounds per contract:

$$N = \frac{120,000}{40,000} = 3$$

A simple matching hedge would involve shorting three contracts.

4. Mini contracts and multipliers

A **mini contract** provides a smaller exposure than a larger version of the product.

For an index future, the contract size is often expressed through a **multiplier**.

If a contract pays \$20 per index point, a 15-point move gives:

$$20 \times 15 = \$300$$

per contract.

5. Delivery location

The exchange specifies where delivery can take place.

Why? A barrel of oil at a convenient refinery location is not economically identical to a barrel thousands of kilometres away.

Transportation and storage costs matter.

Alternative locations may have price adjustments.

6. Delivery months and delivery period

A **delivery month** identifies the contract.

A **delivery period** is the permitted window during which delivery can occur.

Some contracts permit delivery on several dates rather than on one exact day.

7. Price quotation

A **quotation convention** tells you what a displayed number means.

Examples:

- Dollars per barrel.
- Cents per bushel.
- Dollars per ounce.
- Index points.
- Dollars and thirty-seconds for some bond quotations.

Example: cents versus dollars

A corn quote of 650 cents means:

$$\$6.50 \text{ per bushel}$$

For 5,000 bushels:

$$5,000 \times 6.50 = \$32,500$$

A rise of **one cent** creates:

$$5,000 \times 0.01 = \$50$$

of profit for one long contract.

8. Tick and tick value

A **tick** is the minimum permitted price increment.

Tick value converts that increment into money.

If a contract covers 500 units and its tick is \$0.02:

$$\text{Tick value} = 500 \times 0.02 = \$10$$

9. Daily price limits

A **daily price limit** restricts permitted price movements or trading under the exchange's rules.

- **Limit up:** price reaches the upper boundary.
- **Limit down:** price reaches the lower boundary.
- **Limit move:** a movement to one of those boundaries.

Such rules may restrict trading, change trading conditions, or involve halts depending on the contract.

A price limit does not guarantee a maximum total loss. Prices can move again later, and you may be unable to exit when you want.

10. Position limits

Position limits restrict how large a position a participant can hold under applicable rules.

Their purpose includes preventing excessive concentration and manipulation.

Who chooses between permitted delivery alternatives?

In many contracts, the **short** chooses among permitted grades, locations, or delivery dates.

That flexibility benefits the short.

Consequently, all else equal, it tends to **lower the futures price** compared with an otherwise identical contract where delivery terms are fully fixed.

2.3 - Convergence of futures and spot prices

Convergence means that futures and the relevant spot price approach each other as final settlement approaches.

Under matching conditions:

$$F_T = S_T$$

The comparison must use the right asset quality, location, timing, and settlement reference.

Why should they converge?

Suppose immediate delivery is available and:

- Spot price: \$100.

- Futures delivery price: \$108.
- Relevant costs are negligible.

A trader could:

1. Buy the eligible asset for \$100.
2. Sell the futures at \$108.
3. Deliver the asset.
4. Capture \$8.

This is an **arbitrage opportunity**: a price inconsistency that can be locked in through offsetting transactions.

As traders act:

- Buying supports the spot price.
- Selling puts pressure on the futures price.

The gap narrows.

What if futures is below spot?

Suppose spot is \$100 and futures is \$95 during the delivery period.

People who need the commodity may prefer buying futures and awaiting delivery.

But the reverse arbitrage is not automatically as straightforward:

A long may not be able to force immediate delivery. The short can have timing choices.

Borrowing the commodity, short-selling it, and matching delivery can also be difficult.

This asymmetry is important for Hull's Question 2.12.

Basis

A common definition is:

$$\text{Basis} = S - F$$

If spot is 100 and futures is 103:

$$b = -3$$

For an exactly matched contract at final settlement, basis approaches zero.

A farmer's local selling price can still differ from the exchange reference because of location, quality, or transport.

That difference can create **basis risk**: the hedge and actual exposure do not offset perfectly.

2.4 - The operation of margin accounts

This is the central section of Chapter 2.

Why do margin accounts exist?

Suppose you agree to buy something later for \$100. Its market price falls to \$60.

Your contractual obligation is now unfavorable by \$40.

The other party worries:

“Will you still pay, or will you disappear?”

Margin and regular settlement reduce the amount of unpaid loss that can accumulate.

Initial margin

Initial margin is the performance collateral required when opening a position.

It is not:

- An option premium.
- The full purchase price of the commodity.
- A maximum-loss limit.

Analogy: a financial security deposit supporting your ability to fulfil a promise.

Maintenance margin

Maintenance margin is the customer-account threshold below which replenishment is required in Hull's basic model.

Suppose:

- Initial margin: \$6,000.
- Maintenance margin: \$4,500.

If the account falls to \$4,000, you restore it to \$6,000:

$$\text{Deposit} = 6,000 - 4,000 = \$2,000$$

You do not merely deposit \$500 to reach maintenance.

Margin call

A **margin call** is a demand for additional funds or collateral under the account rules.

If you fail to meet it, the broker can close the position.

If losses exceed the account balance, closing the position does not make the unpaid debt disappear.

Daily settlement and marking to market

Marking to market means valuing a position using current market prices.

For futures, this is accompanied by actual settlement of gains and losses.

For a long:

$$\text{Daily P\&L} = NQ(F_t - F_{t-1})$$

For a short:

$$\text{Daily P\&L} = NQ(F_{t-1} - F_t)$$

On entry day, use the actual trade price as the starting price.

Hull's gold example

The investor buys:

- Two gold contracts.
- 100 ounces each.
- Entry price: \$1,450 per ounce.
- Initial margin: \$12,000 total.
- Maintenance margin: \$9,000 total.

Total quantity:

$$Q = 2 \times 100 = 200 \text{ ounces}$$

At the first settlement, price falls to \$1,441.

$$\text{Loss} = 200(1,441 - 1,450) = -\$1,800$$

Account balance:

$$12,000 - 1,800 = \$10,200$$

There is no maintenance call because \$10,200 exceeds \$9,000.

Later, the book's balance falls to \$7,980:

$$7,980 < 9,000$$

Required top-up:

$$12,000 - 7,980 = \$4,020$$

Depositing cash is not another trading loss

Suppose you deposited \$12,000 initially, later added \$4,020, and ultimately received \$14,000 back.

Your result is:

$$14,000 - (12,000 + 4,020) = -\$2,020$$

A deposit moves cash into the account. It does not itself create an additional loss.

Variation margin: terminology needs care

Variation margin describes payments or collateral transfers associated with changes in market value.

Hull also uses the term for the additional money a customer supplies after a margin call.

Distinguish:

- The **daily settlement loss** on the position.
- The **extra customer deposit** needed to restore the account.

They are connected but not identical events.

Why daily changes produce the correct total

For an unchanged long position:

$$Q(F_1 - F_0) + Q(F_2 - F_1) + \cdots + Q(F_n - F_{n-1})$$

Intermediate prices cancel:

$$\text{Total P\&L} = Q(F_n - F_0)$$

This is a telescoping sum.

Futures price versus contract value

After daily settlement, the accumulated gain or loss has been paid.

In the basic model, the continuing contract is effectively reset at the new settlement price.

Therefore:

- The **futures quote** can be \$1,450.
- The freshly settled contract's **current value** can be zero.

Those are different concepts.

Withdrawing excess margin

Under Hull's customer-account model, amounts above required initial margin can be withdrawn.

If initial margin is \$6,000 and the balance becomes \$7,200, the excess is:

$$7,200 - 6,000 = \$1,200$$

Actual withdrawal conditions depend on the account and other exposures.

What can be posted as collateral?

Eligible securities may satisfy initial-margin requirements.

Examples include government securities.

A **Treasury bill** is a short-term government debt instrument.

Securities are often assigned a **haircut**: only part of their market value counts.

If a security worth \$10,000 has a 10% haircut:

$$\text{Collateral value} = 10,000(1 - 0.10) = \$9,000$$

Futures settlement losses require cash under the chapter's description.

Why margin requirements change

Greater price variability generally means a larger potential loss before a position can be closed.

That can justify higher margin.

Requirements may also vary by position structure. A spread with offsetting exposures may require less collateral than two unrelated positions.

The clearing house and its members

A **clearing house** stands between cleared counterparties and manages their obligations.

The usual chain is:

Customer → broker → clearing member → clearing house.

A broker that is not itself a clearing member uses one.

A **clearing member** is an institution authorized to clear through the clearing house and responsible for meeting its requirements.

Customer margin versus clearing-member margin

Customer account in Hull's model	Clearing-member account in Hull's model
Initial and maintenance levels	Required initial margin plus daily settlement
Replenishment after falling below maintenance	Settlement losses must be paid without the same customer-style buffer
Broker manages the client relationship	Clearing house manages member obligations

A member adding new positions may need more initial margin even if existing positions made money.

Netting

Netting combines eligible opposite exposures.

Hull gives the simplified example of 20 long contracts and 15 short contracts being treated as a net five-contract exposure for a margin calculation.

Actual permitted netting depends on account, segregation, and clearing rules.

Guaranty fund

A **guaranty fund**, or default fund, contains resources contributed to help manage losses if a clearing member defaults.

Initial margin, settlement, closeout procedures, and default resources work together.

The 1987 crash lesson

Hull describes customers suffering losses greater than their margin, and some brokers failing after customers did not pay.

The clearing houses nevertheless met the relevant obligations to winning traders in the example.

The lesson: margin reduces default exposure, but it does not guarantee that every customer or intermediary survives.

2.5 - OTC markets, collateral, and central clearing

OTC means *over the counter*: the trade is arranged outside the conventional exchange-contract framework.

OTC does not automatically mean unregulated or uncollateralized.

Credit risk

Counterparty credit risk is the possibility that someone fails to pay what they owe.

Suppose a derivative is worth \$10 million to your company.

If your counterparty fails, you may not receive that value.

Bilateral clearing

Under **bilateral clearing**, the original parties remain responsible to each other.

They commonly use a **master agreement** covering multiple transactions.

A master agreement can specify:

- How obligations are calculated.
- When transactions can be terminated.
- How amounts are determined after default.
- Which transactions can be netted.

An **ISDA Master Agreement** is a widely used form of derivatives agreement discussed in the book.

Credit support annex: CSA

A **CSA** specifies collateral arrangements associated with the trading relationship.

It addresses matters such as eligible collateral, valuation, and when collateral must be provided.

Two-way versus one-way collateral

Under a **two-way agreement**, either party can have to post collateral when the relationship moves against it.

Example

Today, the transactions become \$500,000 more valuable to A and correspondingly less valuable to B.

Under a simple fully collateralized arrangement, B posts \$500,000 to A.

Under a **one-way agreement**, only one party is required to post. This can leave the other party's obligations unsecured.

Central counterparty: CCP

A **CCP** accepts eligible trades and becomes counterparty to both sides.

If A agreed to buy from B, the CCP takes corresponding obligations toward A and B.

It manages risk through margin, valuation, member requirements, and default procedures.

Central clearing can improve netting and standardize risk management. It also concentrates important responsibilities in the CCP.

Numerical example: bilateral versus central clearing

A company has positions worth:

- **+\$20 million** against Bank A.
- **−\$15 million** against Bank B.
- **−\$25 million** against Bank C.

A positive value means the company has an asset; a negative value means it owes value.

Under separate one-way agreements where only the company posts collateral:

$$15 + 25 = \$40 \text{ million posted}$$

Its positive claim against A does not automatically reduce what it owes B or C.

Through one CCP, assuming the stated netting is allowed:

$$20 - 15 - 25 = -\$20 \text{ million}$$

If initial margin is another \$10 million:

$$20 + 10 = \$30 \text{ million posted}$$

Futures settlement versus OTC collateral

In Hull's distinction:

- A futures variation payment **settles a gain or loss**.
- OTC variation collateral can **secure an outstanding obligation** without finally settling it.

That distinction affects interest treatment.

In the book's model, cash OTC collateral generally earns interest for the provider under the agreement. Futures variation settlement is an actual payment, not an outstanding collateral loan on which the payer continues earning interest.

Actual contractual arrangements can differ.

Collateral does not remove all future exposure

Suppose a bank has:

- A derivative worth +\$10 million.
- \$10 million of cash collateral.

Current unsecured exposure can be close to zero.

But after the counterparty defaults, time passes before the trade is closed.

If its value becomes \$13 million while collateral stays at \$10 million:

$$\text{Potential gap} = 13 - 10 = \$3 \text{ million}$$

This is **closeout-period exposure**.

Business Snapshot 2.2: LTCM

LTCM pursued what Hull calls **convergence arbitrage**.

A simplified version:

- Buy a relatively illiquid bond.
- Short a similar, more liquid bond.
- Expect their prices to move closer together.

Liquidity means the ability to trade promptly without a large price impact.

After Russia's 1998 default, investors placed an unusually high value on liquid instruments - a **flight to quality/liquidity**.

The gap widened:

- The assets LTCM owned fell.
- The assets it had shorted rose.
- Collateral demands arrived on both sides.

Heavy **leverage** - large exposures relative to capital - made the situation harder to survive.

The lesson: two positions that usually offset can fail to offset during stress. A strategy called "arbitrage" may still contain funding, liquidity, and convergence risk.

2.6 - Market quotes

A futures quote contains several different prices and activity measures.

Field	Definition
Open	Opening trading price or opening-price measure
High	Highest traded price during the session
Low	Lowest traded price during the session
Prior settlement	Previous official settlement price
Last	Most recent transaction price
Change	In Hull's table, last minus prior settlement
Volume	Contracts traded over the reporting period
Open interest	Contracts that remain outstanding

Last price is not necessarily settlement price

The **settlement price** is the official price used for calculating settlement and related margin amounts.

It is determined under the exchange's procedure.

The most recent trade may be different.

Hull's gold quotation example

For June gold:

- Prior settlement: \$1,434.30.
- Last trade: \$1,425.30.
- Difference: −\$9 per ounce.
- Contract size: 100 ounces.

If that last price becomes the settlement:

$$\text{Long daily loss} = 100(-9) = -\$900$$

The short gains \$900.

Volume versus open interest

This distinction is frequently tested.

Volume counts trades during a period.

Open interest counts outstanding contracts.

One outstanding contract has a long and a short but is counted once.

Buyer's action	Seller's action	Open-interest change
Opens a long	Opens a short	+1
Closes a short	Closes a long	-1
Opens a long	Closes a long	0
Closes a short	Opens a short	0

Each one-contract transaction adds one to volume.

Example

1. A buys from B; both open positions. Open interest becomes 1.
 2. A sells to C; A exits and C replaces A. Open interest stays 1.
 3. C sells to B; both remaining positions close. Open interest becomes 0.
- Total volume is three, but final open interest is zero.

Patterns of futures prices

A **futures curve** shows prices for different maturities at the same time.

Normal market:

Maturity	Price
Near month	100
Three months	103
Six months	106

Prices increase with maturity.

Inverted market: prices decrease with maturity.

A curve can also have a mixed shape.

You may encounter **contango** and **backwardation**, but check the comparison being made. An observed curve slope and a comparison with the expected future spot price are not identical concepts.

The futures price is not simply a forecast. Financing, storage, ownership benefits, and risk pricing can affect it.

2.7 - Delivery

Most futures positions are closed before delivery, but the delivery mechanism still matters.

Notice of intention to deliver

For the physically delivered contracts described by Hull, the short initiates a **notice of intention to deliver**.

It identifies relevant details such as quantity, location, and permitted grade.

The clearing system assigns delivery to an eligible long under its rules.

That long need not be the original trading partner.

Warehouse receipt

A **warehouse receipt** documents the right to specified goods held in a warehouse.

Taking commodity delivery can mean receiving this document rather than watching a truck arrive immediately.

Storage costs and other responsibilities then follow the contract's arrangements.

Three dates to distinguish

Date	Meaning
First notice day	First day a delivery notice can be submitted
Last notice day	Last day a delivery notice can be submitted
Last trading day	Last day the contract can be traded

They are not automatically the same.

A long seeking to avoid delivery should close before the relevant notice exposure, taking account of broker deadlines.

Why delivery may use the latest settlement price

Suppose you entered long at 100 and the final delivery invoice uses 110.

You have already received total futures gains of 10.

Therefore:

$$\text{Effective purchase cost} = 110 - 10 = 100$$

You should not expect both:

- Delivery at the original 100.
- An additional 10 of settlement gains.

That would count the same benefit twice.

Cash settlement

A **cash-settled contract** finishes through a payment based on a specified reference value.

Stock indices are natural examples because delivering the complete index portfolio can be inconvenient.

Example

An index futures contract has a \$50-per-point multiplier.

Entry: 4,000. Final settlement: 4,020.

$$\text{Total long gain} = 50(4,020 - 4,000) = \$1,000$$

If \$800 has already been settled on earlier days, the final remaining adjustment is \$200.

2.8 - Types of traders and types of orders

Trader roles

Term	Meaning
Futures commission merchant - FCM	Intermediary handling customer futures business
Local	Own-account trader in the book's historical floor terminology
Hedger	Reduces a pre-existing exposure
Speculator	Deliberately takes market exposure
Arbitrageur	Exploits a price inconsistency through offsetting transactions
Scalper	Seeks small profits over very short holding periods
Day trader	Normally closes positions within the trading day
Position trader	Holds positions over longer periods

These classify different things.

For example, someone can simultaneously be a **local**, a **speculator**, and a **day trader**.

Market order

An instruction to trade promptly at available prices.

Example: "Buy one contract now."

It prioritizes execution, not a guaranteed price.

Limit order

Sets the worst acceptable execution price.

- Buy limit at 100: pay 100 or less.
- Sell limit at 100: receive 100 or more.

It may never execute.

Stop order

A stop activates after its trigger condition is reached.

Example

You are long, and the current price is 105.

You place a sell stop at 100.

If the market falls to the trigger, the instruction becomes a market-type sell order.

The stop price is not a guaranteed execution price.

If the market gaps to 95, execution may occur around 95.

Stop-limit order

Combines:

1. A stop trigger.
2. A limit on the permitted execution price.

Example

- Sell stop: 100.
- Sell limit: 99.

After triggering, sell only at 99 or higher.

If the market drops directly to 95, the order may remain unfilled.

Stop-and-limit

Hull uses this term when the stop price and limit price are the same.

Market-if-touched - MIT

An MIT order becomes market-type after a specified favorable price is touched.

For a long position:

- Sell stop below market: exit after an adverse fall.
- Sell MIT above market: take profit after a favorable rise.

Neither guarantees the exact trigger price as the final execution price.

Discretionary or market-not-held order

Gives the broker specified discretion, often over timing, in an attempt to obtain a better execution.

Time conditions

Instruction	Meaning
Day order	Expires at the end of the trading day
Time-of-day order	Active only during a specified interval
Good-till-cancelled / open order	Remains subject to cancellation and venue expiry rules
Fill-or-kill	Execute the complete quantity immediately or cancel

A limit controls price. A time condition controls how long or under what timing conditions the instruction remains active.

2.9 - Regulation

Hull discusses the historical US framework, particularly:

- **CFTC:** Commodity Futures Trading Commission.
- **NFA:** National Futures Association.
- The expansion of derivatives oversight following the financial crisis.

The chapter describes functions such as reporting, registration, financial requirements, customer protection, investigation, and disciplinary action.

These explain the system's purposes; the edition's wording is not a current legal operating manual.

Systemic risk

Systemic risk is the danger that problems at one institution spread and disrupt the wider financial system.

Example

A cannot pay B. B consequently struggles to pay C. Several institutions sell assets urgently, depressing prices and causing further losses.

Clearing, collateral, and reporting seek to manage parts of this problem.

Cornering the market

A participant attempts to control enough deliverable supply while holding a large long futures position.

Shorts approaching delivery can struggle to obtain the commodity and may have to buy back contracts at high prices.

The chapter discusses responses such as:

- Higher margin.
- Tighter position limits.
- Restrictions on increasing positions.

- Required closeouts.

Front running

Front running means exploiting knowledge of a customer's order to trade first for oneself.

Example

A trader knows a large customer purchase is about to be executed. The trader buys first for their own benefit, then sells after the customer's order moves the price.

Hull also discusses historical allegations of overcharging customers and withholding sale proceeds.

2.10 - Accounting and tax

The key is to distinguish **three different clocks**:

1. When the position gains or loses economically.
2. When cash is paid.
3. When accounting or tax recognizes the result.

Accounting example from Hull

A company buys corn futures:

- Entry: 650 cents per bushel.
- Year-end: 670 cents.
- Exit next February: 680 cents.
- Quantity: 5,000 bushels.

Total gain:

$$5,000(6.80 - 6.50) = \$1,500$$

The first-year change is:

$$5,000(6.70 - 6.50) = \$1,000$$

The second-year change is:

$$5,000(6.80 - 6.70) = \$500$$

Mark-to-market accounting

In the chapter's basic nonhedge example, the annual changes are recognized when they occur:

- \$1,000 in the first year.
- \$500 in the second.

Hedge accounting

Hedge accounting is special accounting treatment for qualifying hedging relationships.

Its purpose includes aligning recognition of the hedge's effects with the relevant hedged item.

If the futures hedges a qualifying February purchase, Hull's simplified example matches the \$1,500 gain with that later transaction.

The futures cash flows still settle daily. Accounting treatment does not postpone the actual margin payments.

Fair value and balance sheet

Fair value is the accounting measurement of an instrument's current value under the relevant framework.

A **balance sheet** reports assets, liabilities, and equity.

The text discusses recognizing derivatives at fair value and imposing documentation and effectiveness requirements for hedge accounting.

It refers to standards such as **FAS 133** and **IAS 39** in its historical setting.

An economic hedge does not automatically qualify for special accounting treatment.

Tax terminology

Term	Meaning
Ordinary income	Income under the ordinary classification, including relevant business results
Capital gain/loss	Gain or loss classified under capital-asset rules
Short-term / long-term	Tax classifications associated with holding-period or special rules
Recognition	Including an amount in the relevant taxable period
Carryback	Applying a qualifying loss to an earlier tax year
Carryforward	Applying a qualifying loss to a later tax year

The 60/40 rule described in the text

For the relevant US futures tax treatment discussed by Hull:

- 60% of the gain/loss receives long-term capital classification.
- 40% receives short-term classification.

For a \$1,000 gain:

\$600 long-term classification

\$400 short-term classification

These are not tax rates.

The text also describes year-end treatment as though applicable positions were closed at their market values.

Business hedges

Properly identified qualifying business hedges generally receive ordinary treatment with recognition matched to the hedged income or expense in the chapter's framework.

Accounting and tax definitions of a hedge are not identical.

For exercises, calculate annual gains first, then apply the stated hedge/speculator classification. Do not assume the old US tax discussion applies to Morocco.

2.11 - Forward versus futures contracts

Feature	Forward	Futures
Arrangement	Usually private/OTC	Exchange-standardized
Terms	Can be customized	Defined by the contract
Settlement timing	Basic model settles at maturity	Daily gains/losses are settled
Delivery timing	Often one specified date	May involve a delivery window
Typical exit	Termination, offsetting arrangement, or maturity	Opposite trade in the same contract
Credit protection	Depends on collateral and clearing arrangements	Clearing and margin structure

Same total nominal gain, different cash-flow timing

Hull considers £1 million exposure:

- Entry rate: \$1.5000 per pound.
- Final rate: \$1.7000 per pound.

The gain is:

$$1,000,000(1.7000 - 1.5000) = \$200,000$$

A simple forward delivers that result at maturity.

Futures delivers a sequence of gains and losses whose sum is \$200,000, ignoring interest.

Why timing matters

Suppose prices fall sharply before recovering.

The long futures trader pays cash during the fall, possibly needing a loan.

The forward trader does not face those same daily settlement payments in the simple uncollateralized model.

Thus:

Equal total cash amounts can have different economic values because they arrive at different times.

Funding rates and the price path matter.

Foreign exchange quotations

The chapter warns that futures and forward quotes may use reciprocal conventions.

Suppose a forward quote is:

1.1000 CHF/USD

This means:

1 USD = 1.10 CHF

Convert it to dollars per Swiss franc:

$\frac{1}{1.10} = 0.90909$ USD/CHF

If the futures quote is:

0.9000 USD/CHF

someone selling Swiss francs receives more dollars under the quoted forward, before costs and cash-flow-timing differences.

Never compare 1.1000 directly with 0.9000 without checking units.

The extra statistical idea in Question 2.35

The chapter's final exercise asks how much margin might be needed to withstand an adverse two-day move.

The relevant terms are:

- **Daily price change:** $F(t) - F(t-1)$.
- **Standard deviation:** a measure of how widely those changes vary.
- **Normal distribution assumption:** a model for the shape of possible changes.
- **99% one-sided threshold:** a loss level exceeded with approximately 1% probability under that model.
- **Backtesting:** comparing a model's predictions with historical outcomes.

If daily changes have standard deviation s , and you assume independent, stable, normally distributed changes:

$$\text{Two-day standard deviation} = \sqrt{2} s$$

For quantity Q , a 99% one-sided normal loss threshold is:

$$M \approx Q \times 2.326 \times \sqrt{2} \times s$$

The factor 2.326 is the standard normal 99th percentile.

This is a model-based estimate, not a guarantee. Large jumps, changing volatility, and dependence can make actual losses exceed the model's predictions more frequently.

The distinctions you should be able to explain aloud

Common confusion	Correct distinction
Margin versus premium	Margin supports obligations; premium purchases an option right
Initial versus maintenance margin	Opening requirement versus replenishment trigger
Margin call versus trading loss	A call requests funding; the price movement creates the loss
Futures price versus contract value	Quoted reference price versus current value of remaining obligations
Last versus settlement price	Latest trade versus official settlement reference
Volume versus open interest	Trades during a period versus outstanding contracts
Closing versus rolling	Remove the position versus move exposure to another maturity
First notice versus last trading day	Delivery exposure can begin before trading ends
Stop versus limit	Triggered execution instruction versus execution-price restriction
Stop versus put protection	A stop can fill poorly; a put creates a contractual right
Collateral versus final settlement	Security for a claim versus payment discharging an obligation
Cash settlement versus accounting recognition	When money moves versus when the result is recorded
Hedging versus making money on the derivative	A hedge is assessed together with the original exposure
Notional versus credit exposure	Reference size versus the amount actually exposed to default
Central clearing versus no risk	Clearing manages risks; it does not eliminate every risk

A strong answer to almost any Chapter 2 question identifies the position, the units, the contractual obligation, the cash-flow dates, and what happens if someone cannot pay.